

Preface

Topology

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Chapter 1

Topological Spaces

§ 1.1. Topological Spaces

Definition 1.1.1. A topology on a set X is a collection T of subsets of X such that:

- (1) $\emptyset, X \in T$;
- (2) if $\{U_i\}_{i \in I} \subseteq T$, then $\bigcup_{i \in I} U_i \in T$;
- (3) if $\{U_i\}_{i=1}^n \subseteq T$, then $\bigcap_{i=1}^n U_i \in T$.

We call the pair (X, T) a topological space.

Example 1.1.1. Let $X = \{a, b, c\}$. Then: $T_1 := \{\emptyset, \{a, b, c\}, \{b\}, \{a, b\}, \{b, c\}\}$ and $T_2 := \{\emptyset, \{b\}, \{a, b, c\}\}$ are topologies on X . However, $T_3 = \{\emptyset, \{a\}, \{b\}, \{a, b, c\}\}$ is *not* a topology on X .

Example 1.1.2 (Every set admits a topology). Let X be a set. The *discrete topology* on X is $T = \mathcal{P}(X)$. The *trivial topology* on X is $T = \{\emptyset, X\}$.

Example 1.1.3. Let $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$ be vectors in $\mathbb{R}^n$. Define $d(\vec{x}, \vec{y}) := \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$. This is a metric on $\mathbb{R}^n$, and we will see this induces a topology on $\mathbb{R}^n$.

Definition 1.1.2. Let (X, d) be a metric space, $x \in X$, and $\epsilon > 0$. The open ball centered at x with radius ϵ is denoted $B(x, \epsilon) = \{y \in X \mid d(x, y) < \epsilon\}$.

Definition 1.1.3. Let (X, d) be a metric space. A set $U \subseteq X$ is open if for all $x \in U$, there exists $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$. The collection of open sets of X is denoted τ_X .

Proposition 1.1.1. Let (X, d) be a metric space. Then (X, τ_X) is a topological space.

Proof. Clearly $\emptyset, X \in \tau_X$. Let $\{U_i\}_{i \in I} \subseteq X$ be an arbitrary family of open sets. Let $x \in \bigcup_{i \in I} U_i$. Then $x \in U_i$ for some i . So there exists an $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U_i \subseteq \bigcup_{i \in I} U_i$. Now let $\{U_i\}_{i=1}^n$ be a finite collection of open sets. Let $x \in \bigcap_{i=1}^n U_i$. Then $x \in U_i$ for all $i = 1, \dots, n$. So there exists $\epsilon_i > 0$ such that $B(x, \epsilon_i) \subseteq U_i$ for all $i = 1, \dots, n$. Define $\epsilon := \min\{\epsilon_1, \dots, \epsilon_n\}$. Then $B(x, \epsilon) \subseteq U_i$ for all $i = 1, \dots, n$. Hence $B(x, \epsilon) \subseteq \bigcap_{i=1}^n U_i$. Thus (X, τ_X) is a topological space. $\square$

Example 1.1.4. Using the same terminology from Example 1.1.3, since $d(\vec{x}, \vec{y})$ is a metric, $(\mathbb{R}^n, \tau_{\mathbb{R}^n})$ is a topological space.

§ 1.2. Topologies Generated by a Basis

Definition 1.2.1. Let X be a set. A basis for a topology on X is a collection $\mathcal{B}$ of subsets of X such that:

- (1) for all $x \in X$, there exists $B \in \mathcal{B}$ such that $x \in B$.
- (2) If $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Remark. Let (X, d) be a metric space and $\mathcal{B}$ a basis for a topology on X . It follows from the above definition that for any $U \in \tau_X$, there exists an open basis element $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Example 1.2.1. The sets $\mathcal{B}_1 := \{(a, b) \mid a < b\}$ and $\mathcal{B}_2 := \{(a, b) \mid a < b, a, b \in \mathbb{Q}\}$ are bases for a topology on $\mathbb{R}$. These definitions extend nicely to $\mathbb{R}^n$ by considering open balls.

Example 1.2.2 (Every set admits a basis for a topology). Let X be a set. Then $\mathcal{B} := \{\{p\} \mid p \in X\}$ is a basis for a topology on X .

Proposition 1.2.1. Let X be a set and $\mathcal{B}$ a basis for a topology on X . Let $\mathcal{T}$ be the set containing the empty set and every subset of X which is equal to a union of basis elements of $\mathcal{B}$. Then $(X, \mathcal{T})$ is a topological space.

Proof.

□

Definition 1.2.2. Let X be a set and $\mathcal{B}$ a topology on X . The topology defined in Proposition 1.2.1 is called the topology generated by $\mathcal{B}$.

Example 1.2.3. From Example 1.2.1, $\mathcal{B}_1$ and $\mathcal{B}_2$ generate the standard topology on $\mathbb{R}^n$. From Example 1.2.2, $\mathcal{B}$ generates the discrete topology on X .

§ 1.3. Product Topologies

Definition 1.3.1. Let $(X, \mathcal{T}_1)$ and $(Y, \mathcal{T}_2)$ be topological spaces. The product topology on $X \times Y$ is the topology generated by the basis $\mathcal{B} := \{U \times V \mid U \in \mathcal{T}_1, V \in \mathcal{T}_2\}$.

Remark. Note that $\mathcal{B}$ is not necessarily a topology on $X \times Y$. Indeed, consider $X = Y = \{0, 1\}$ in the discrete topology. Then the sets $\{0\} \times \{0\}$ and $\{0\} \times \{1\}$ are in $\mathcal{B}$ as per the above definition, but their union $\{(0, 0), (0, 1)\}$ is not. Hence $\mathcal{B}$ is not a topology.